

INDIAN INSTITUTE OF TECHNOLOGY, GANDHINAGAR
ELECTRICAL ENGINEERING DEPARTMENT

EE 322: Analog and Mixed Signal Circuits
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Project Proposal

By

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1. Introduction:

The primary challenge in analog synthesis is recreating the complex sonic character of an acoustic instrument, such as a snare drum, using electronic circuits. Unlike a simple tone, a snare drum's sound is a composite of two distinct elements: the resonant, pitched "thump" from the drum's body and the sharp, noisy "snap" from the metal snares rattling against the bottom drumhead. The core issue, therefore, is to design separate circuit blocks that can generate each of these sounds and then combine them in a balanced and musically pleasing way.

This process is fundamental to the design of classic analog drum machines, like the iconic Roland TR-808, whose sounds have shaped entire genres of music. By breaking down the acoustic properties of a snare into its basic components—a decaying sine wave for the body and filtered white noise for the snares—engineers and hobbyists can build a circuit from the ground up. This exploration is not just an academic exercise in electronics; it is the foundational practice behind creating the very instruments that define electronic music.

2. Problem Statement:

The problem is to design and implement a purely analog electronic circuit that synthesizes a convincing and musically versatile snare drum sound. This breaks down into three specific technical challenges:

1. **Synthesizing the Drum Body:** The circuit must generate a pitched, resonant tone that emulates the sound of a drumstick hitting the membrane. This involves creating a stable oscillator, typically a "pinged" resonant filter like a Bridged-T network, that produces a decaying sine wave when excited by a trigger pulse. Key parameters to control are the fundamental pitch (frequency) and the decay time of the sound.
2. **Synthesizing the Snare Wires:** The circuit must produce the sharp, atonal "snap" of the metal wires. This requires a white noise source that is subsequently shaped into a percussive burst using a Voltage Controlled Amplifier (VCA) and a fast envelope generator. Crucially, this noise signal must be aggressively high-pass filtered to remove low frequencies, ensuring it sounds bright and crisp rather than muddy.

3. **Mixing and Control:** The final challenge is to seamlessly mix the "body" and "snare" signals into a single cohesive output. The design must also incorporate voltage control (CV) over its primary parameters, allowing for external modulation from a sequencer or other modules to create dynamic and expressive rhythmic patterns.

Scope and Significance

The scope of this project is strictly limited to the domain of analog electronics. The solution must be realized with discrete components like op-amps, transistors, resistors, and capacitors, without relying on digital signal processing (DSP) or microcontrollers. The final circuit should be a self-contained module that accepts a standard gate/trigger signal as an input and provides a line-level audio signal as its output.

The project's significance is twofold. First, it serves as a profound educational tool, providing a practical application for a wide range of fundamental analog circuit theories. Second, it holds significant artistic relevance. The ability to build and modify classic synthesizer circuits empowers musicians and artists to move beyond presets and samples, allowing them to craft unique sonic signatures and contribute to the vibrant culture of DIY electronic music.

Impact of a Solution

Solving this problem is critical for achieving a sonically authentic and musically useful instrument. A failure to properly balance the two core components will result in a sound that is either a simple electronic tom or an unmusical burst of static. A successful solution, however, provides more than just a single sound; it creates a versatile percussive voice that can be tweaked from tight, snappy funk snares to long, booming snares used in electronic dance music.

The potential impact of a well-designed, open-source snare circuit is significant for the electronic music community. It provides artists, students, and hobbyists with the knowledge and a proven template to build their own hardware. This not only makes custom instrumentation more accessible but also fosters innovation, as builders experiment with and modify the core design, leading to the creation of entirely new sounds and musical possibilities.

3. Proposed Solution:

a. Overview of Approach:

The proposed method for creating the analog snare drum circuit is a modular, bottom-up engineering approach. This strategy involves deconstructing the complex target sound into its fundamental components and developing a dedicated electronic sub-circuit for each, ensuring that every part is functional before integrating it into the final system.

The approach begins by treating the two core sonic elements—the pitched drum body and the noisy snare snap—as separate design problems. Each will be tackled independently, starting with theoretical design and validation through SPICE software simulation. This "virtual prototyping" step allows for rapid optimization of component values and circuit topology, minimizing errors before physical assembly.

Following a successful simulation, each functional block will be built and tested in isolation on a solderless breadboard. Using an oscilloscope and audio monitoring, the real-world performance will be compared against the simulated results. Once every individual module is verified, they will be integrated into the complete circuit. The final stage will involve an iterative process of system-level tuning, where the overall sound is refined through a combination of objective measurement and, most importantly, subjective listening tests to achieve a musically expressive and versatile instrument.

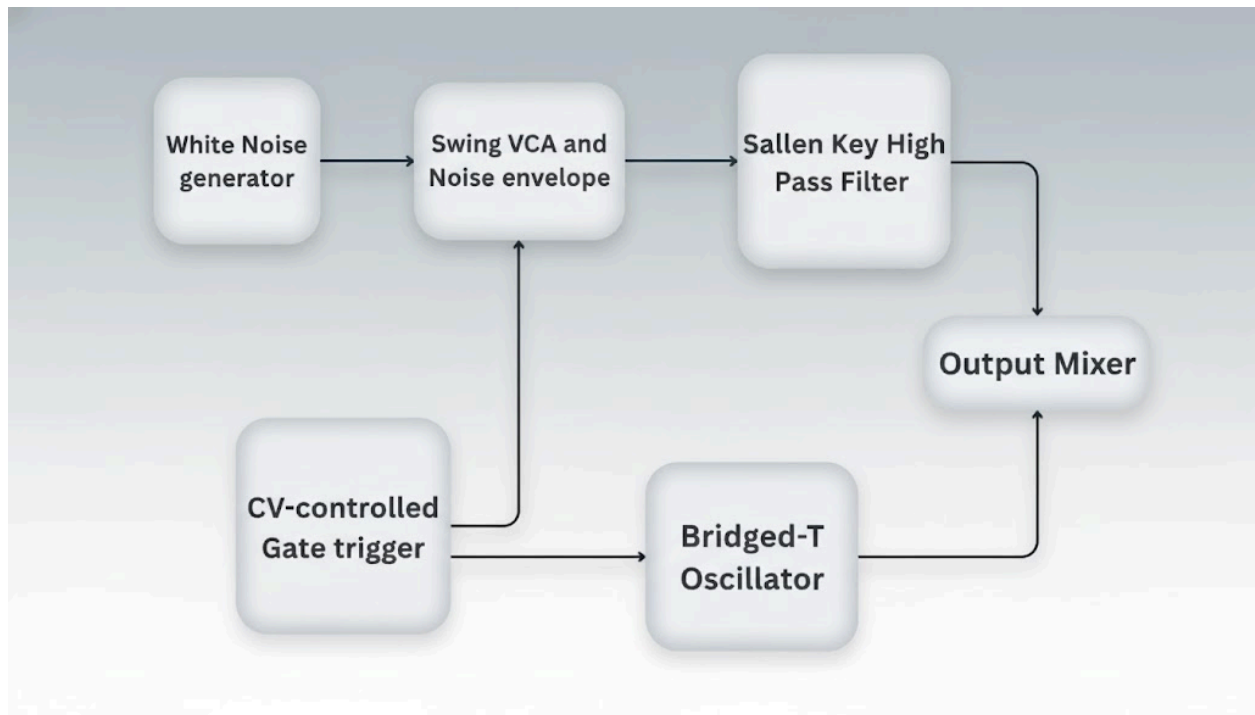
b. Components and Blocks:

- **CV Controlled Gate to Trigger:** This circuit, a CV Controlled Gate-to-Trigger Converter, is a fundamental block in analog synthesis designed to transform a sustained gate signal into a brief, percussive trigger pulse. Its operation begins when the rising edge of the input gate passes through a capacitor-resistor network, which generates a sharp positive voltage spike. A diode immediately clips any corresponding negative spike from the gate's falling edge, ensuring only a single pulse is created. This raw spike is then fed into an op-amp that shapes and amplifies it into a clean, robust trigger. Finally, the "ACCENT CV" input provides dynamic control by allowing an external voltage to adjust the amplitude of this output pulse, making it possible to create emphasized or "accented" notes in a rhythmic sequence.
- **Bridged T oscillator:** This Bridged-T Oscillator circuit is designed to generate a pitched, percussive sound, emulating the resonant body of a drum. Instead of oscillating continuously, it functions as a high-resonance filter that is "pinged" by an incoming trigger pulse, causing it to ring out and naturally decay like a struck object. The fundamental pitch of this ringing is adjusted by the "TUNE" control, while the "DECAY" knob controls a positive feedback loop to determine how long the sound sustains before fading to silence. Additionally, the "ATTACK" circuit adds a punchy, percussive quality by creating a rapid pitch envelope at the very beginning of the sound, resulting in a complete synthesized drum voice that outputs a decaying sine wave.
- **White Noise Generator (Snare Wires Source):** This block is responsible for creating the raw, unpitched audio signal that forms the metallic "snap" of the snare wires. Its function begins by using a **reverse-biased NPN transistor** to induce Zener breakdown, which generates chaotic, randomly fluctuating currents. the signal is then **AC coupled** to remove DC offset and fed into a **high-gain non-inverting op-amp stage** for necessary amplification before being passed to the shaping circuits
- **Swing VCA & Noise Envelope:** This composite block transforms the continuous white noise into a sharp, percussive event with a controllable decay time(controlled by the potentiometer). The timing is governed by an **envelope generator**, which is created when the main trigger signal charges a large capacitor via a diode, resulting in a fast attack and a regulated discharge. The resulting voltage modulates a simple **transistor-based amplifier** (the VCA core) to shape the noise's volume over time. A key control, the **Snare Decay CV potentiometer**, is placed across the capacitor's discharge path to

regulate the envelope's fall time.

- **Sallen-Key High-Pass Filter:** Crucial for achieving a clean, crisp snare sound, this active filter is implemented to remove the low-frequency content from the shaped noise signal. The Sallen-Key topology employs two passive RC filter stages coupled with an **active op-amp element**. This design introduces positive feedback to generate **resonance (Q)** at the cutoff frequency, which intentionally boosts the high-frequency "snap" while preventing any low-end rumble to the output.
- **Output Mixer:** This is the final stage of the circuit, responsible for creating a single, cohesive audio signal. It uses an op-amp configured as an **Inverting Summing Amplifier** to combine the two separately generated audio paths: the decaying sine wave from the Bridged-T Oscillator (Drum Body) and the filtered noise from the Snare Wires.

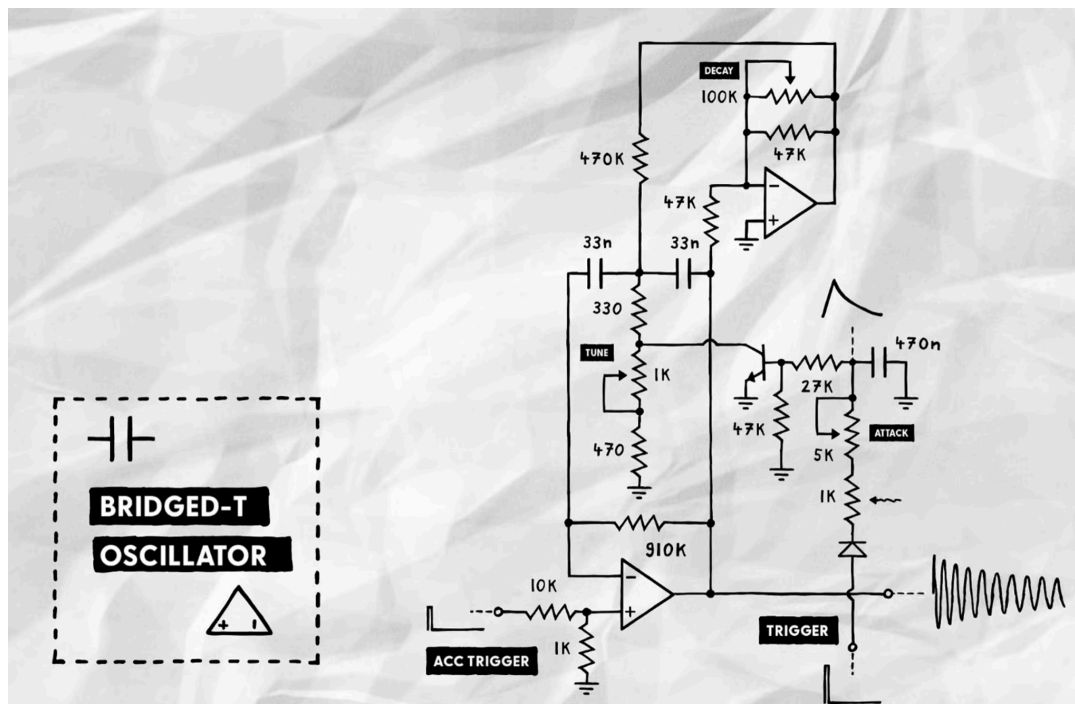
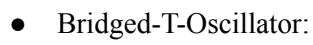
- **Block Diagram:**



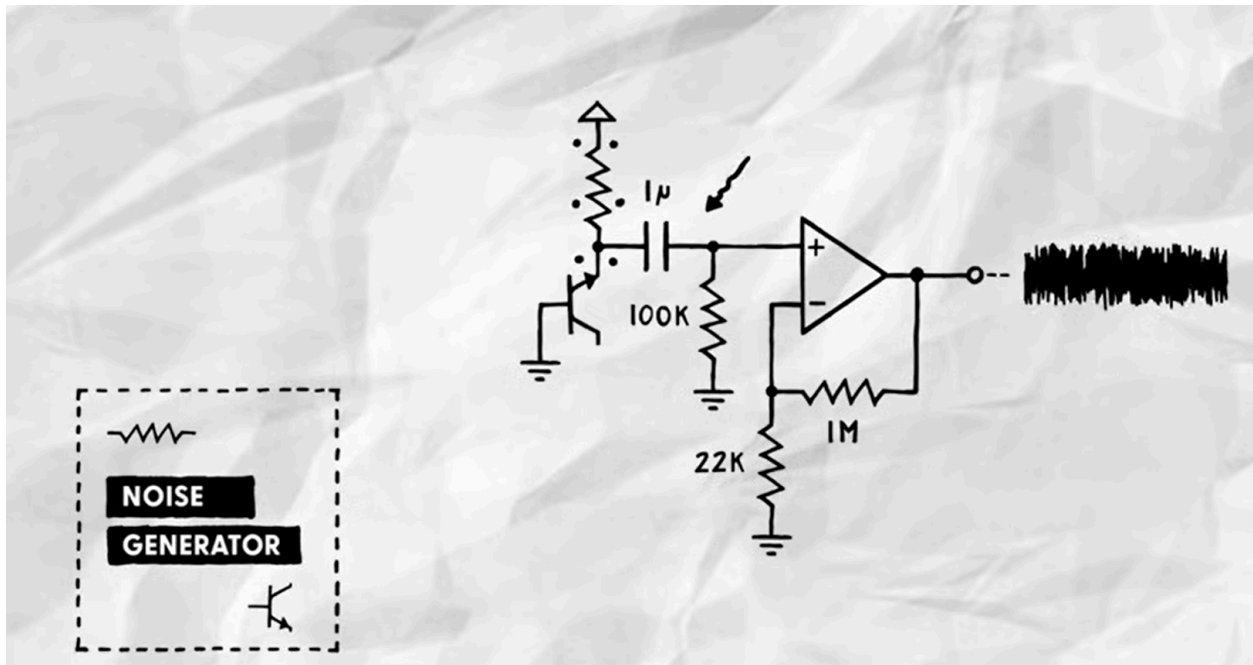
4. Working of the circuit:

a. Circuit Diagrams:

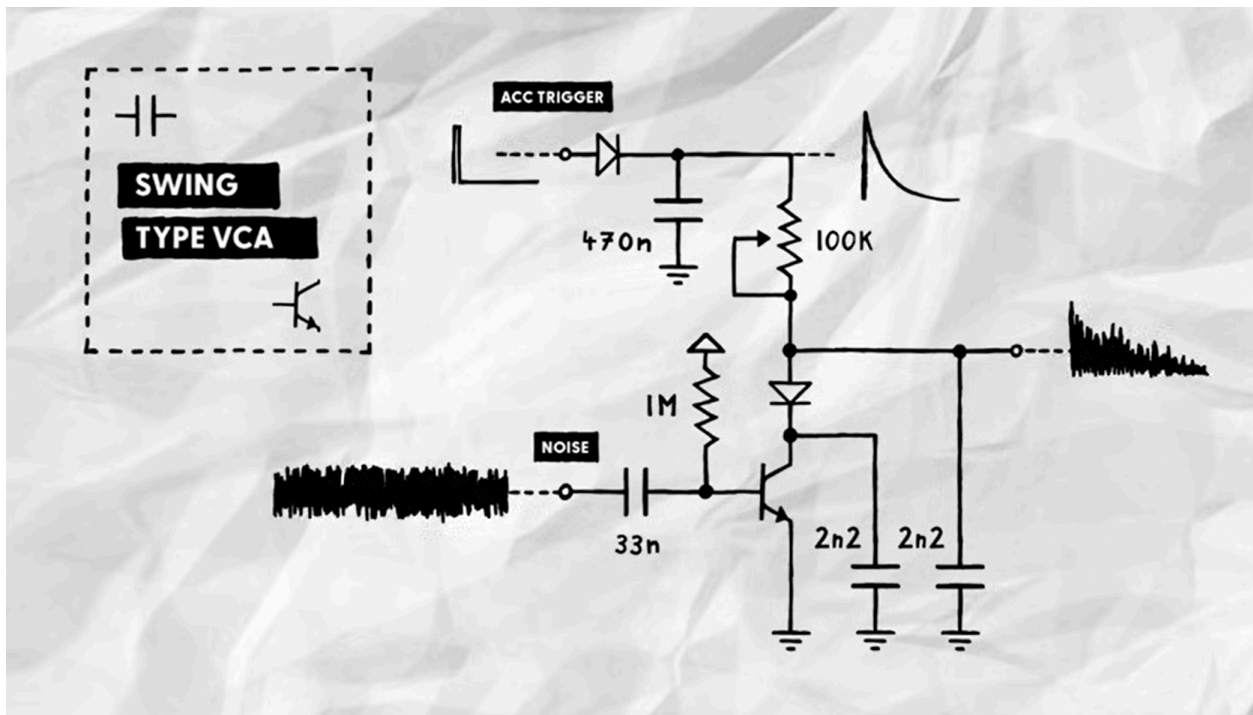
- CV Controlled Gate-to-Trigger:



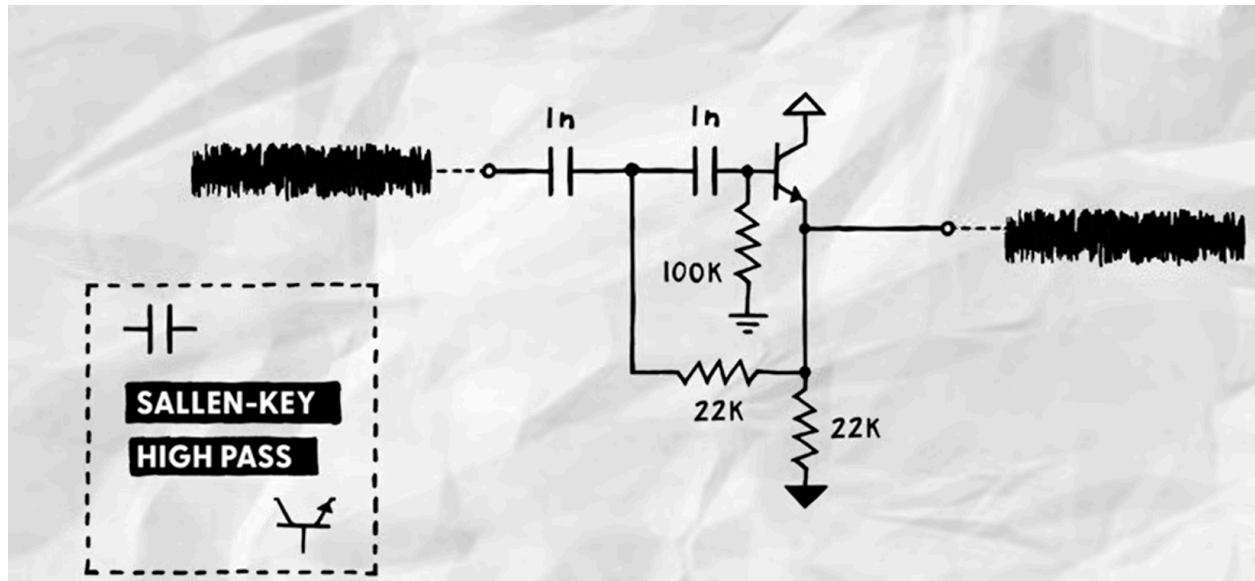
- Noise generator:-



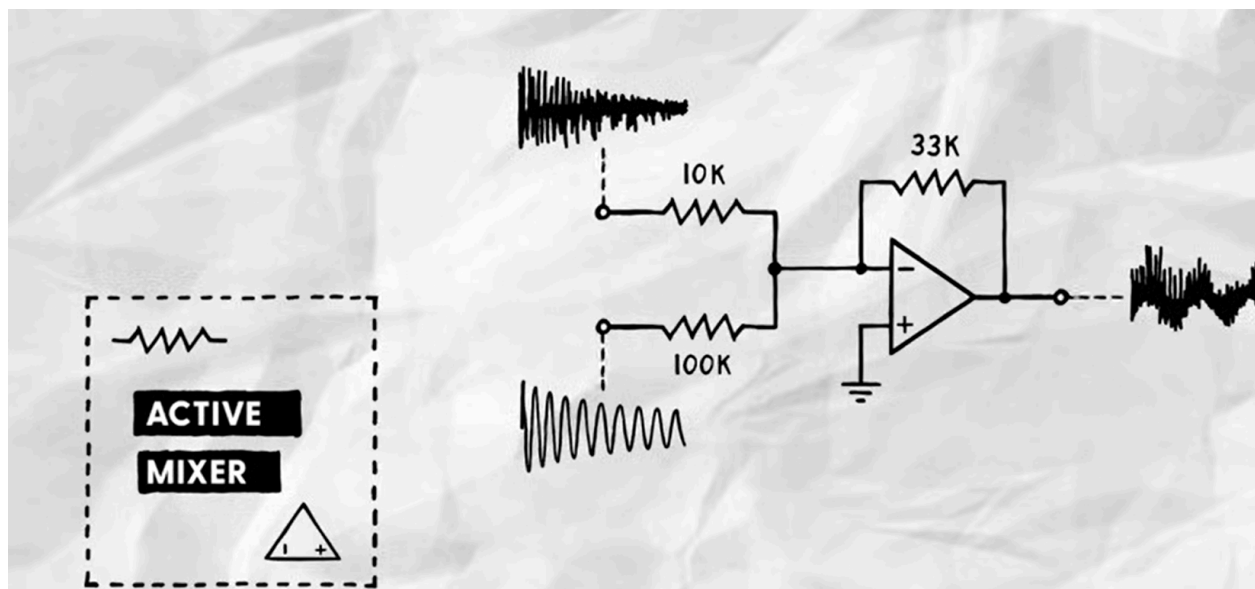
- Swing VCA & Noise Envelope:



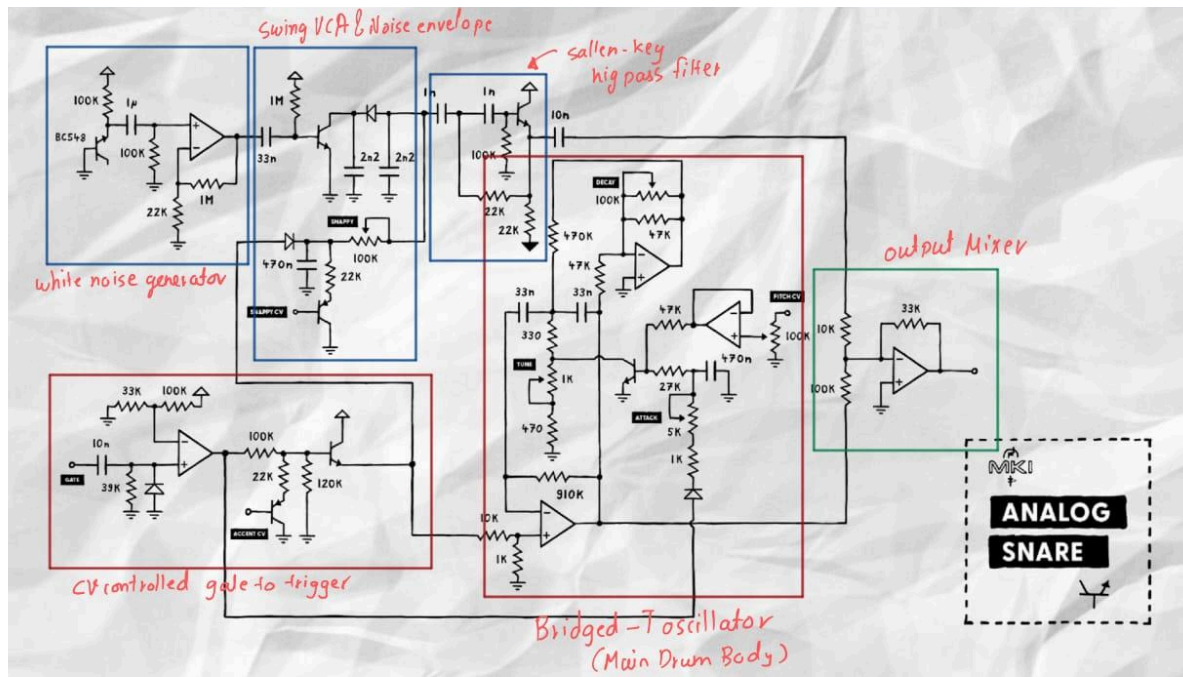
- Sallen-Key High-Pass Filter:



- mixer :-



- **Final Circuit**



- **CV Controlled Gate-to-Trigger:**

1. Set Up Power and the Op-Amp

First, connect your power supply's +V, -V, and Ground rails to the power rails on your breadboard. Place the op-amp IC onto the breadboard, straddling the center divider. Connect its power pins to the corresponding +V and -V rails.

2. Build the Input Differentiator

This stage turns the gate's "edge" into a spike.

Connect the input signal from your function generator (a square wave) to one end of the 10nF capacitor. From the other end of the capacitor, connect the 39K resistor to the ground rail. Connect the diode in parallel with the 39K resistor, with its cathode (the end with the stripe) connected to ground.

Test: Connect an oscilloscope probe to the point where the capacitor, resistor, and diode meet. You should see a sharp positive spike for every rising edge of your input square wave.

3. Add the Op-Amp Shaper

This stage cleans up the spike into a usable pulse.

Connect the spike signal from the previous step to the non-inverting (+) input of the op-amp. Create the feedback loop: connect the 33K resistor from the inverting (-) input to ground. Then, connect the 100K resistor from the inverting (-) input to the op-amp's output pin.

Test: Connect the oscilloscope probe to the op-amp's output. You should now see a clean, rectangular trigger pulse that corresponds to the initial spike.

4. Construct the Output Stage

This stage buffers the output and incorporates the accent control.

Connect the op-amp's output through the other 100K resistor to the base of the main output transistor.

Connect the 120K resistor from the transistor's base to ground. Connect this transistor's collector to the +V rail. The final trigger signal is taken from its emitter. Now, add the accent control transistor. Connect its emitter to ground. Connect its base to your "ACCENT CV" input point. Connect its collector through the 22K resistor to the base of the main output transistor.

Test: View the final output from the main transistor's emitter on your oscilloscope. With 0V at the "ACCENT CV" input, the trigger should be at its full height. As you apply a positive DC voltage to the Accent CV input, you should see the trigger's amplitude get smaller

● **Bridged-T-Oscillator:**

1. Build the Trigger Input Comparator

This stage takes an external trigger and creates the internal "ping" that excites the circuit.

Create a voltage divider for the "ACC TRIGGER" input using the 10K and 1K resistors. Connect the midpoint of these resistors to the non-inverting (+) input of the first op-amp. Connect the output of this op-amp to the main "TRIGGER" line that the rest of the circuit will use.

Test: Feed a pulse from your function generator into the "ACC TRIGGER" input. On your oscilloscope, you should see a clean, sharp pulse at the op-amp's output.

2. Construct the Bridged-T Resonator Core

This is the heart of the circuit that creates the ringing sound. It can be tricky, so take your time.

From the main "TRIGGER" line (output of the first op-amp), connect the 910K resistor to a new node. Let's call this the "T-Junction". From the T-Junction, connect the two 33nF capacitors in series. The point between these two capacitors is your main audio output point. From the other side of the capacitors, build the resistive leg to ground: connect the 330Ω resistor, the 1K TUNE pot, and the 470Ω resistor in series to the ground rail. The circuit won't do much yet, but you've built the core resonant structure.

3. Add the Decay Feedback Amplifier

This stage takes the weakly ringing signal and uses positive feedback to make it sustain.

Connect the audio output point (between the two 33nF caps) to the non-inverting (+) input of the second op-amp. Connect the output of the second op-amp back to its inverting (-) input through the 100K

DECAY pot and a 47K resistor in series. Also connect a 47K resistor from the inverting (-) input to ground. Now, connect the output of this second op-amp back to the audio output point through another 47K resistor.

Test: The circuit should now be fully functional. Connect the audio output to your oscilloscope and amplifier. When you send in a trigger, you should see and hear a decaying sine wave. Adjusting the TUNE pot will change its pitch, and the DECAY pot will change how long it rings out.

4. Integrate the Attack Circuit

This final stage adds the initial punchy pitch-sweep to the sound.

From the main "TRIGGER" line, build the envelope generator: connect the 1K resistor and the diode in series to a new node. From this node, connect the 470nF capacitor and a 47K resistor to ground.

Also from this node, connect the 2.7K resistor and the 5K ATTACK pot in series to the base of the NPN transistor. Finally, connect the transistor into the resonator circuit: connect its emitter to ground and its collector to the top of the resistive leg (the point between the 330Ω resistor and the TUNE pot).

Test: Trigger the sound again. Now, as you turn the ATTACK pot, you should hear the initial character of the sound change from a soft "boop" to a sharp, percussive "pew!", as the pitch quickly sweeps at the beginning.

● **White Noise Generator:**

1. Set Up Power and Transistor Core (Reverse Biasing):

Place a low-noise NPN transistor (BC547) onto the breadboard. Connect the transistor's base and collector together, and tie this junction to the Ground rail. Connect the emitter to the positive power rail (+V, +12V) through a high-value resistor (68KΩ) to bias the base-emitter junction into Zener breakdown, which generates the raw noise.

2. AC Coupling (DC Blocking)

The raw noise is very quiet and contains a large DC offset. To isolate the audio, connect a small capacitor (10nF) from the transistor's emitter (the noise source) to a new node. Connect a resistor (10KΩ) from this new node to Ground to complete the AC coupling and center the noise signal around 0V.

3. High-Gain Amplification

Place an op-amp IC and connect its power pins. Feed the AC-coupled noise signal (from the capacitor/resistor junction in step 2) into the non-inverting (+) input of the op-amp. Configure the op-amp as a high-gain non-inverting amplifier (e.g., using a 1MΩ feedback resistor and a 22KΩ resistor to ground) to amplify the quiet noise to a usable voltage level.

4. Test

Connect the op-amp output to an oscilloscope or an audio amplifier. You should observe a continuous, amplified, and erratic signal centered around 0V, confirming the creation of a steady, amplified white noise signal.

- **Swing VCA & Noise Envelope:**

1. Build the Envelope Generator

Connect the main TRIGGER pulse (from the Gate-to-Trigger circuit) through a diode (e.g., 1N4148) to a new node. Connect a large capacitor (e.g., 1 μ F) from this node to ground. This sets the initial fast attack time.

2. Add Variable Snare Decay

From the same capacitor node, connect the Snare Decay CV potentiometer (e.g., 100K Ω) in series with a fixed resistor (e.g., 22K Ω) to create the discharge path. The output voltage across this resistor will serve as the control voltage for the VCA.

3. Construct the VCA Core

Set up a transistor (e.g., NPN) as a simple amplifier. Feed the White Noise Generator's output into the transistor's base via an appropriate biasing network. The collector circuit will include a resistor network that serves as the VCA control point.

4. Connect Envelope to VCA

Connect the control voltage from the Snare Decay CV potentiometer (from step 2) to the VCA's control point (e.g., the collector resistor junction). This applies the decaying voltage to the VCA, ensuring the noise signal's volume tracks the envelope's shape.

5. Test:

Connect the VCA output to an amplifier. When triggered, the continuous noise should be gated into a short, percussive burst. Adjusting the decay potentiometer should visibly and audibly change the length of the noise burst, simulating the tightness of the snare wires.

- **Sallen-Key High-Pass Filter:**

1. Set Up the Op-Amp

Place an op-amp and connect its power pins. The filter will be configured as an active filter with a gain greater than unity to achieve resonance.

2. Construct the Passive Filter Stages

Connect the output of the Swing VCA through two series capacitors (1nF each) to the non-inverting (+) input of the op-amp. Connect a resistor (100K Ω) from the junction between both capacitors to Ground. Connect a second resistor (22K Ω) from the op-amp's non-inverting (+) input to Ground.

3. Add Resonance (Feedback)

Connect a feedback resistor ($22\text{K}\Omega$) from the op-amp's output back to its inverting (-) input. Connect a small resistor ($10\text{K}\Omega$) from the inverting (-) input to Ground. The ratio of these resistors sets the filter's gain, which introduces and controls the amount of resonance (Q) at the cutoff frequency.

4. Test

Pass the shaped noise burst through the filter. Observe the output on an oscilloscope or speaker. The low-frequency rumble should be aggressively attenuated, and the high frequencies should be emphasized with a metallic, resonant "bite."

● **Output Mixer:**

1. Set Up Inverting Summing Amplifier:

Place the final op-amp on the breadboard and connect its power. Connect its non-inverting (+) input directly to the Ground rail.

2. Connect Drum Body Input:

Connect the audio output from the Bridged-T Oscillator (the decaying sine wave) through its dedicated input resistor ($100\text{K}\Omega$) to the op-amp's inverting (-) input.

3. Connect Filtered Noise Input

Connect the audio output from the Sallen-Key High-Pass Filter through its dedicated input resistor ($10\text{K}\Omega$) to the same inverting (-) input. The ratio of these resistors ($100\text{K}\Omega$ vs $10\text{K}\Omega$) sets the initial volume balance.

4. Set Final Gain:

Connect a feedback resistor ($33\text{K}\Omega$) from the op-amp's output back to the inverting (-) input. This resistor sets the overall system output gain to a line-level standard.

5. Test:

Apply the trigger to the full system. The final output from this op-amp should be the completed snare drum sound, confirming that the pitched body and filtered noise are seamlessly combined and balanced.

5. Brief Explanation of the division of work among the three of you.

- Ayush Umare (23110346):- CV Controlled Gate-to-Trigger and Output mixer
- Yash Sonone (23110318) :- Bridged-T Oscillator
- Sumedh Wankhede (23110364):- White noise generator, Swing VCA and noise envelope and Sallen key high pass filter.

6. References:

[1] "Designing a TR-808 style snare drum from scratch," *www.youtube.com*.
https://www.youtube.com/watch?v=hULEn2_4Unw

[2] Igor Brichkov, "Force Sensitive Piezo Triggered Analog Snare Drum Circuit From Scratch," *You Tube*, Feb. 11, 2025. <https://www.youtube.com/watch?v=LeoYNhAllds> (accessed Oct. 18, 2025).